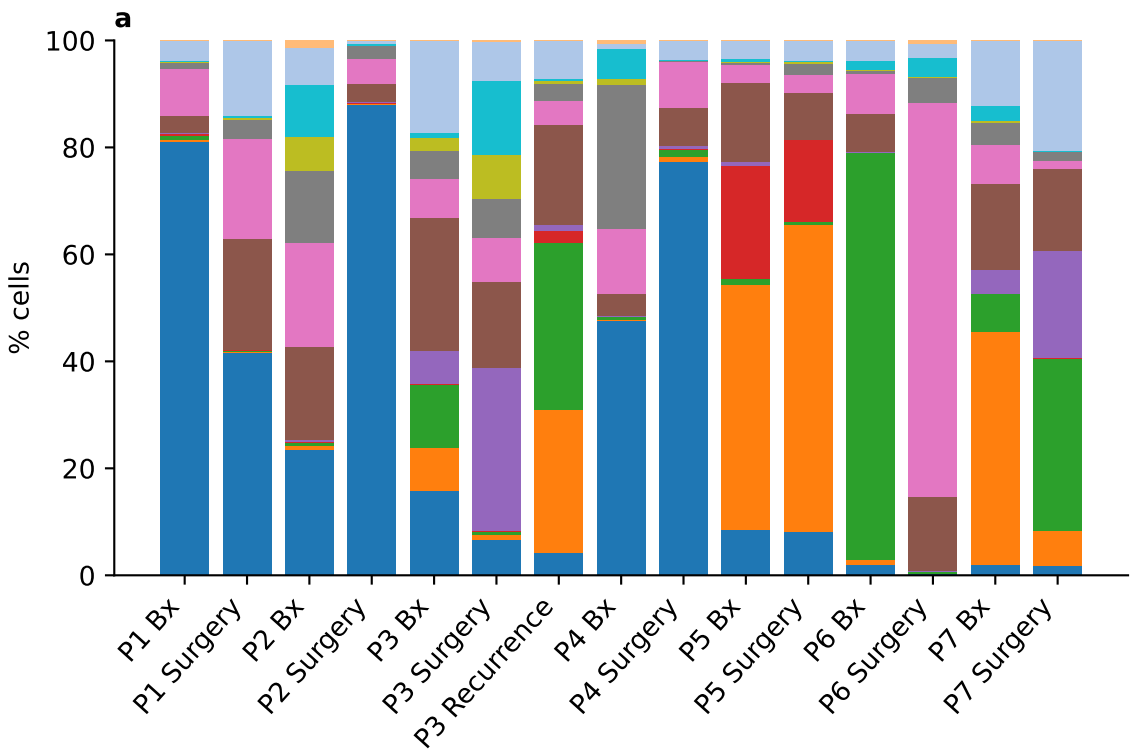
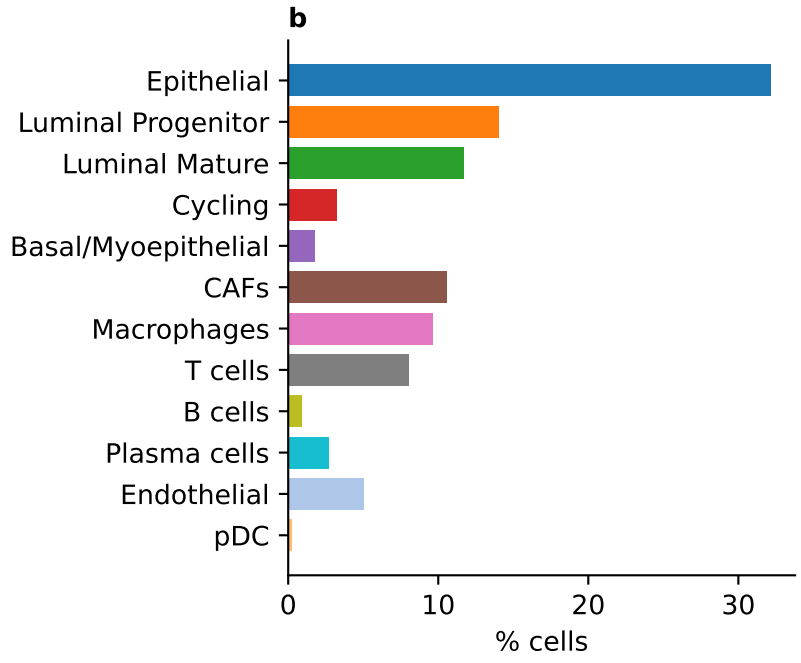
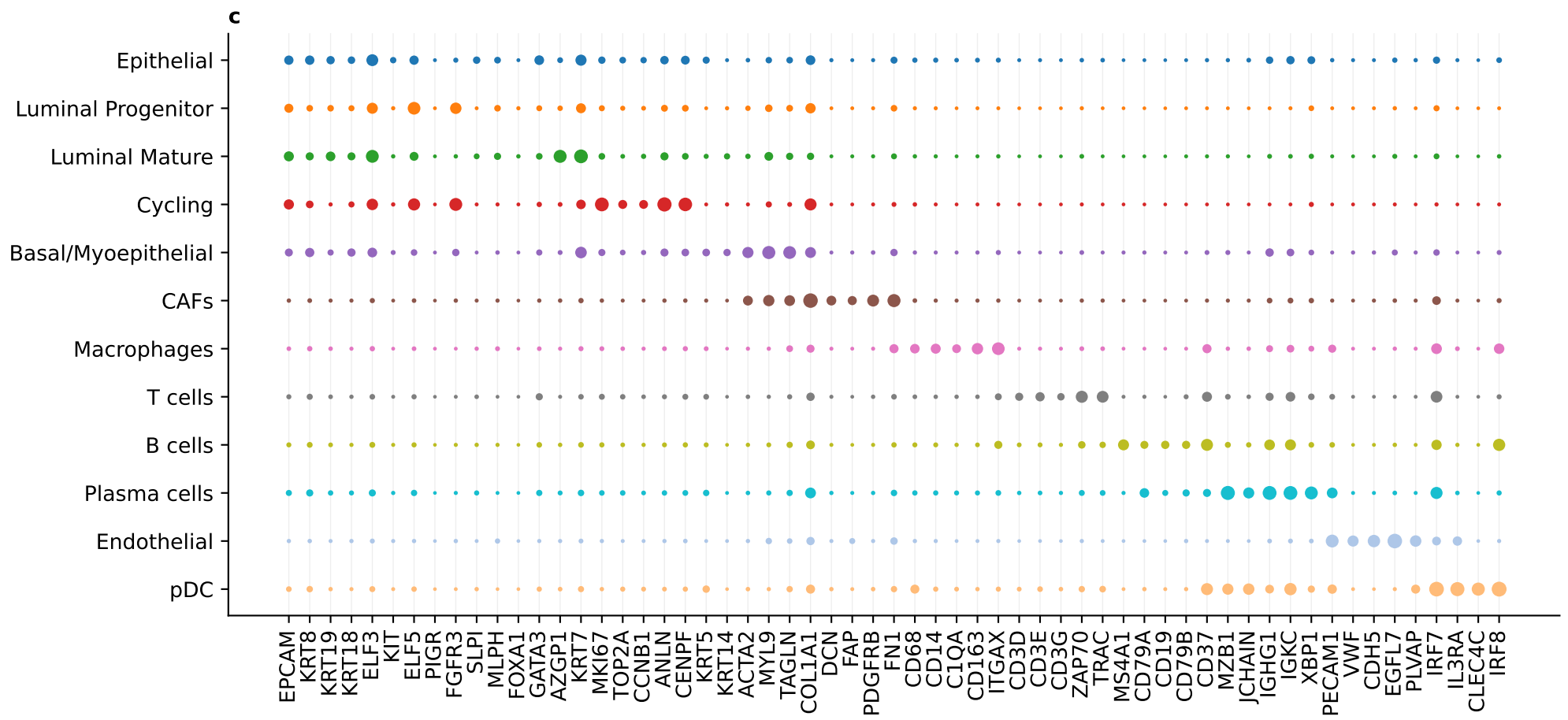
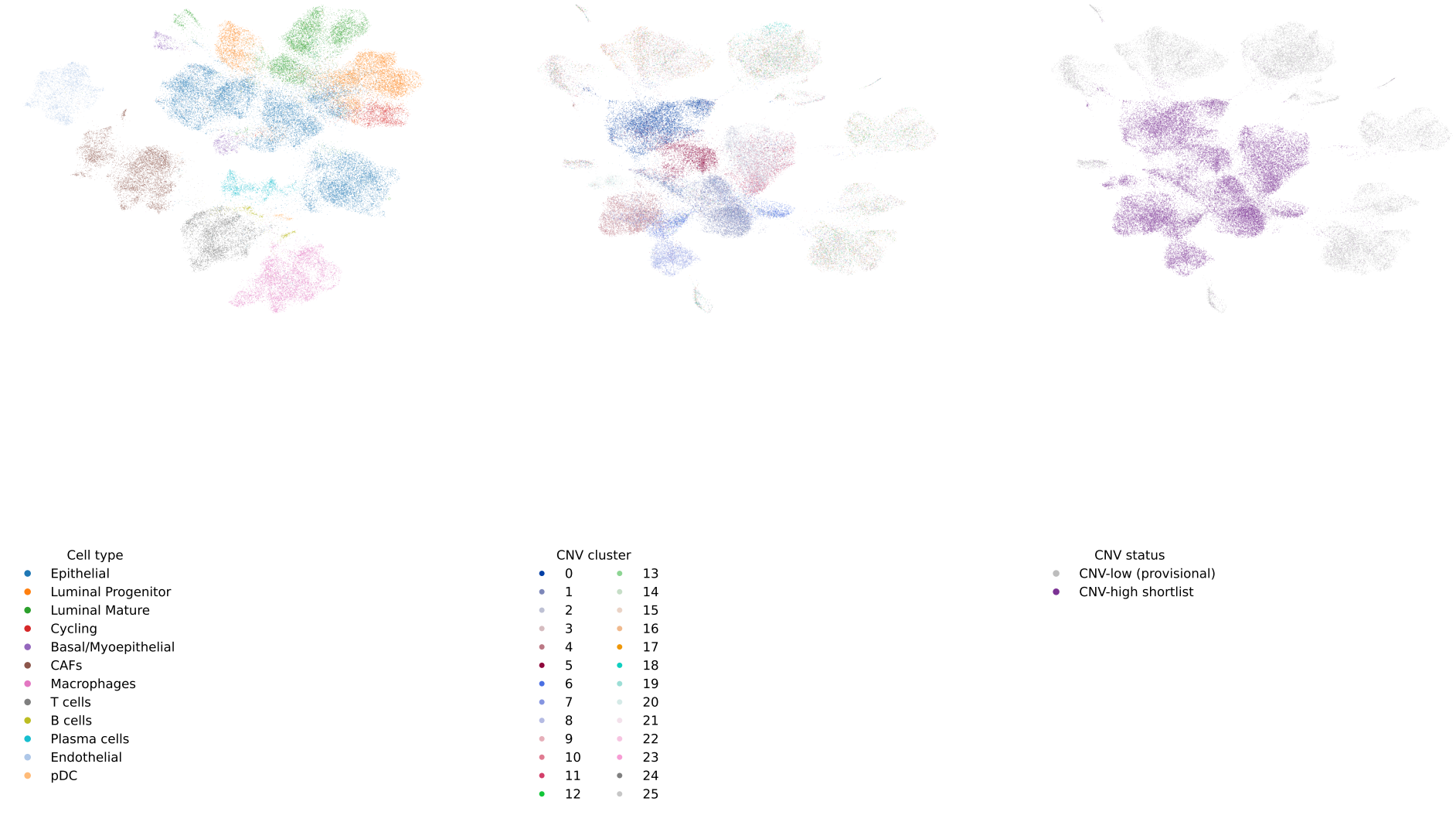
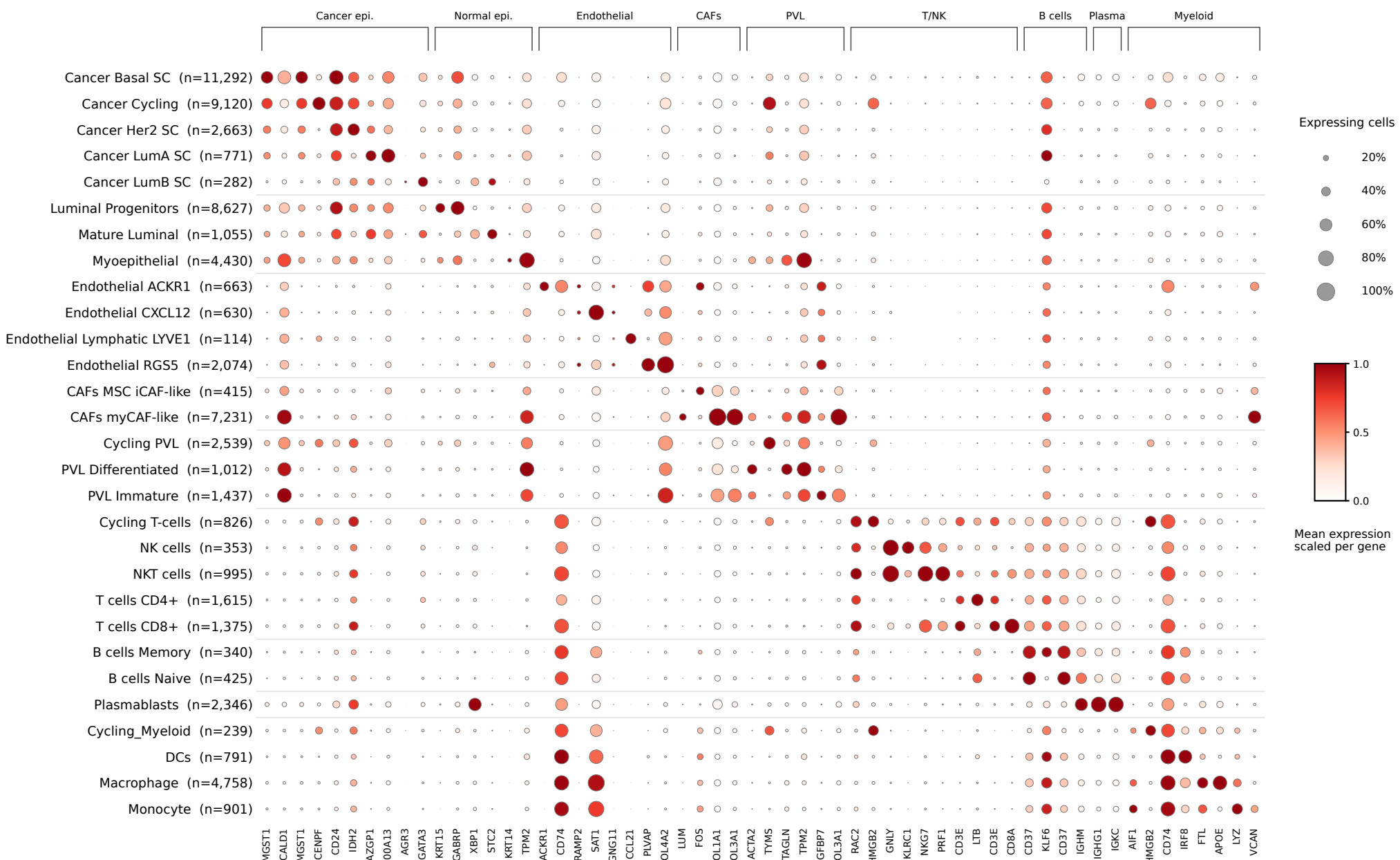

a
100
80
60
40
20
0
% cells
P1 Bx
P1 Surgery
P2 Bx
P2 Surgery
P3 Bx
P3 Surgery
P3 Recurrence
P4 Bx
P4 Surgery
P5 Bx
P5 Surgery
P6 Bx
P6 Surgery
P7 Bx
P7 Surgery


b
Epithelial
Luminal Progenitor
Luminal Mature
Cycling
Basal/Myoepithelial
CAFs
Macrophages
T cells
B cells
Plasma cells
Endothelial
pDC
0
10
20
30
% cells


c
Epithelial
Luminal Progenitor
Luminal Mature
Cycling
Basal/Myoepithelial
CAFs
Macrophages
T cells
B cells
Plasma cells
Endothelial
pDC
EPCAM
KRT8
KRT19
KRT18
ELF3
KIT
ELF5
PIGR
FGFR3
SLPI
MLPH
FOXA1
GATA3
AZGP1
KRT7
MKI67
TOP2A
CCNB1
ANLN
CENPF
KRT5
KRT14
ACTA2
MYL9
TAGLN
COL1A1
DCN
FAP
PDGFRB
FN1
CD68
CD14
C1QA
CD163
ITGAX
CD3D
CD3E
CD3G
ZAP70
TRAC
MS4A1
CD79A
CD19
CD79B
CD37
MZB1
JCHAIN
IGHG1
IGKC
XBP1
PECAM1
VWF
CDH5
EGFL7
PLVAP
IRF7
IL3RA
CLEC4C
IRF8


d
celltype
e
cnv_leiden
f
cnv_status
Cell type
Epithelial
Luminal Progenitor
Luminal Mature
Cycling
Basal/Myoepithelial
CAFs
Macrophages
T cells
B cells
Plasma cells
Endothelial
pDC
CNV cluster
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
CNV status
CNV-low (provisional)
CNV-high shortlist

g Reference-derived marker expression across all 29 full-Wu minor identities



Two saved Wu reference-derived markers per identity; shared genes may repeat. Dot size: cells with expression >0. Color: mean log1p(10k counts/cell), including zeros, scaled per gene.